```
}
}
}

public class TrafficDemo{
public static void main(String[] args){
TrafficSignal t=new TrafficSignal();
Thread th=new Thread(t);
th.start();
}
}
```

## Slip 16

**1) Write a Java program to display Fibonacci series.**

```
import java.util.Scanner;

public class Fibonacci {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter the number of terms: ");
      int n = sc.nextInt();

      int a = 0, b = 1;

      System.out.println("Fibonacci Series:");

      for (int i = 1; i <= n; i++) {
```

```
        System.out.print(a + " ");

        int c = a + b;
        a = b;
        b = c;
      }

      sc.close();
   }
}
```

**2) Write a program to sort HashMap by keys and display the details before sorting and after sorting.**

```
import java.util.*
public class SortHashMap {
   public static void main(String[] args) {
      HashMap<Integer, String> map = new HashMap<>();
      map.put(3, "C");
      map.put(1, "A");
      map.put(5, "E");
      map.put(2, "B");
      map.put(4, "D");
      System.out.println("HashMap before sorting:");
      for (Map.Entry<Integer, String> entry : map.entrySet()) {
         System.out.println(entry.getKey() + " : " + entry.getValue());
      }
      TreeMap<Integer, String> sortedMap = new TreeMap<>(map);
```

```
        System.out.println("\nHashMap after sorting by keys:");
        for (Map.Entry<Integer, String> entry : sortedMap.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }
}
```

**OR**

**2) Write a Java program to define an interface "Operation" which has Methods, area(), volume(). Define a constant PI having a value of 3.142.**

```
interface Operation {
    double PI = 3.142;
    void area();
    void volume();
}
class Circle implements Operation {
    double radius;
    Circle(double radius) {
        this.radius = radius;
    }
    public void area() {
        System.out.println("Area of Circle = " + (PI * radius * radius));
    }
    public void volume() {
        System.out.println("Volume is not applicable for a circle.");
    }
}
```